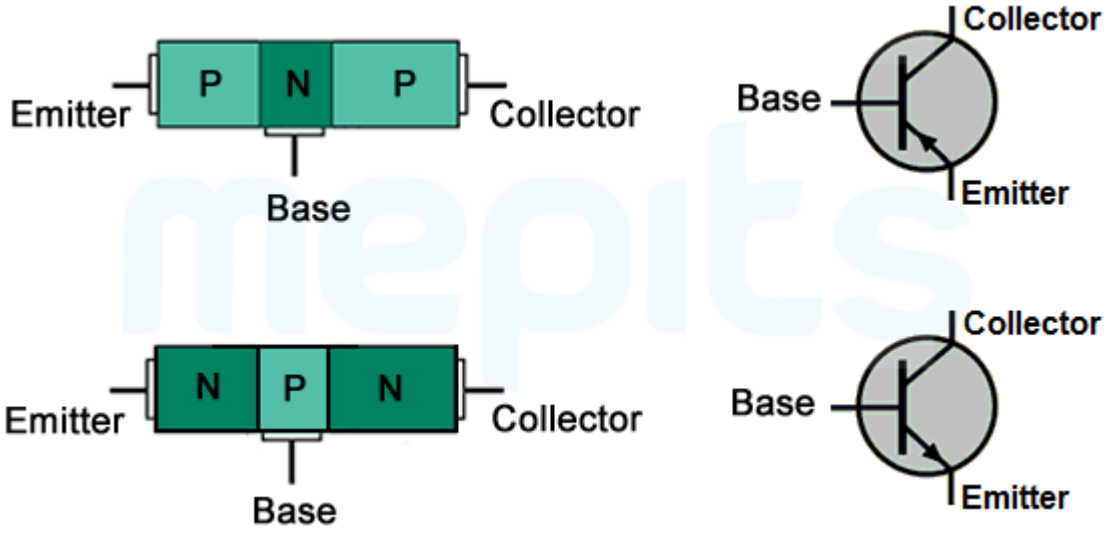
Syllabus: Bipolar Junction Transistors: Transistor Construction, Transistor Operation, Common Base Configuration, Transistor Amplifying Action, Common Emitter Configuration, Common Collector Configuration, DC Biasing – Operating Point, Fixed – Bias Configuration, Emitter – Bias Configuration, Voltage-Divider Bias Configuration. (Text 2: 3.1-3.7, 4.1-4.5)

Module-3 - Bipolar Junction Transistors

The term bipolar reflects the fact that holes and electrons participate in the injection process into the oppositely polarized material. If only one carrier is employed (electron or hole), it is considered a unipolar device

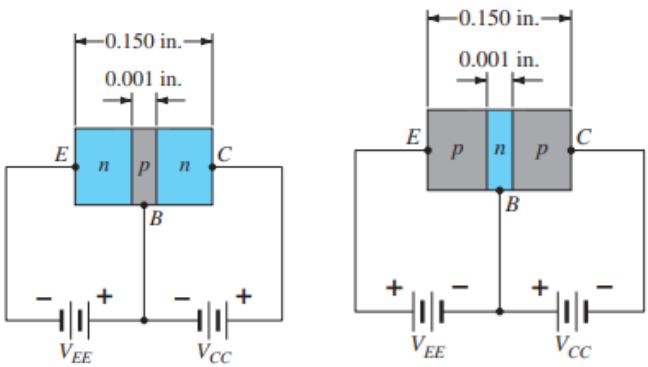
Transistor Construction:

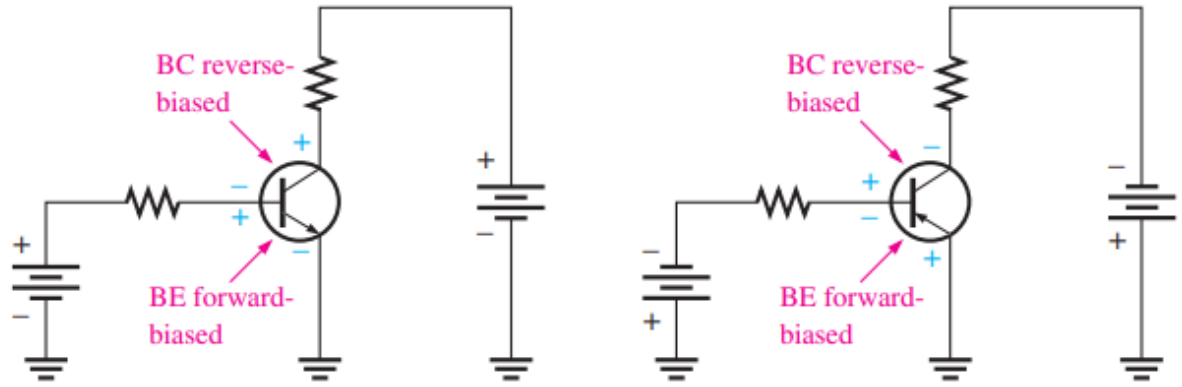


The transistor is a three-layer semiconductor device consisting of either two n - and one p -type layers of material or two p - and one n -type layers of material. The former is called an npn and pnp transistor respectively.

The emitter layer is heavily doped, with the base and collector only lightly doped. The outer layers have widths much greater than the sandwiched p - or n -type material.

Transistor Operation:



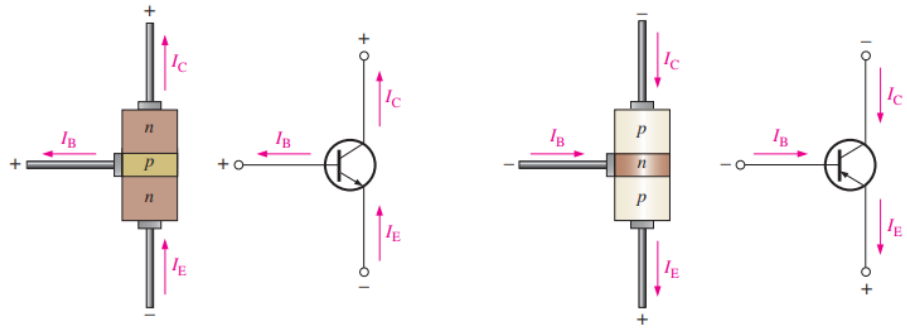


(a) *npn* (b) *pnp*

In both cases the base-emitter (BE) junction is forward-biased and the base-collector (BC) junction is reverse-biased. This condition is called *forward-reverse bias.*

The heavily doped n-type emitter region has a very high density of conduction-band (free) electrons. These free electrons easily diffuse through the forward based BE junction into the lightly doped and very thin p-type base region. A small percentage of the total number of free electrons injected into the base region recombine with holes and move as valence electrons through the base region and into the emitter region as hole current. When the electrons that have recombined with holes as valence electrons leave the crystalline structure of the base, they become free electrons in the metallic base lead and produce the external base current. Most of the free electrons that have entered the base do not recombine with holes because the base is very thin. As the free electrons move toward the reverse-biased BC junction, they are swept across into the collector region by the attraction of the positive collector supply voltage. The free electrons move through the collector region, into the external circuit, and then return into the emitter region along with the base current. The emitter current is slightly greater than the collector current because of the small base current that splits off from the total current injected into the base region from the emitter.

<u>Transistor Currents:</u>



(a) *npn* (b) *pnp*

$I_E = I_C + I_B$

The collector current, however, comprises two components—the majority and the minority carriers. The minority-current component is called the leakage current and is given the symbol I_{CO} (I_C current with emitter terminal Open). The collector current, therefore, is determined in total by

$I_C = I_{C\ majority} + I_{C0\ minority}$

Characteristics of Transistor:

- **Input Characteristics:** The curve describes the changes in the values of input current with respect to the values of input voltage, keeping the output voltage constant.
- **Output Characteristics:** The curve is obtained by plotting the output current against output voltage, keeping the input current constant.
- **Current Transfer Characteristics:** This characteristic curve describes the variation of output current in accordance with the input current, keeping the output voltage constant.

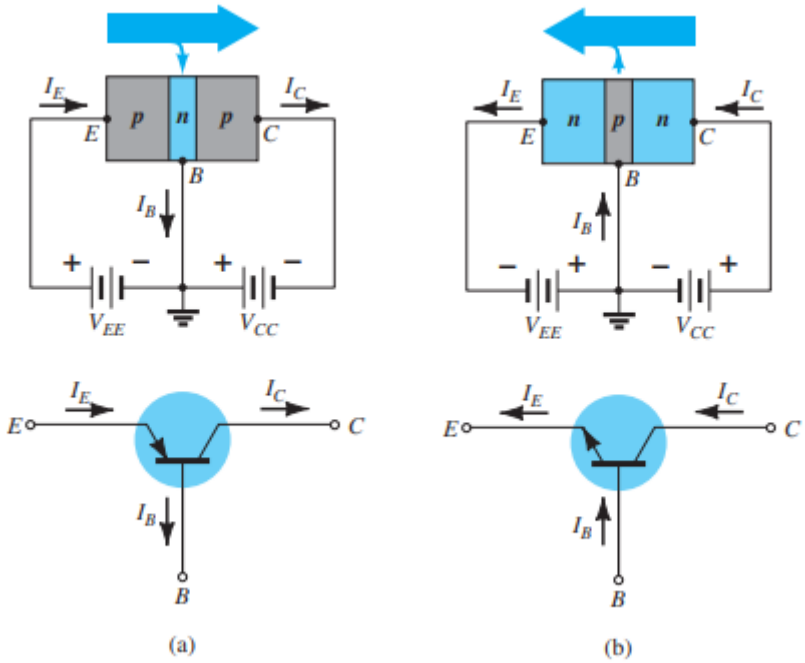
Configuration of Transistor:

The three types of transistor circuit configurations are:

- Common Emitter Transistor
- Common Base Transistor
- Common Collector Transistor(emitter follower).

Common Base Configuration:

The common-base terminology is derived from the fact that the base is common to both the input and output sides of the configuration. In addition, the base is usually the terminal closest to, or at, ground potential.



In the Common base circuit for NPN and PNP the input is given between emitter and base terminals and output is taken from collector and base terminals. The input voltage is denoted as V_{BE} and the output voltage is denoted as V_{CE}. In all the configuration the base emitter junction is always forward biased and the collector base junction is reverse biased.

In the common base configuration of NPN circuit emitter is N type base is of P type and collector is of N type. The emitter base terminals are forward biased so the majority charge carriers in the emitter that is the electrons gets repelled by the negative applied voltage and in the same way the majority charge carriers in the base that is the holes gets repelled by the positive applied voltage.

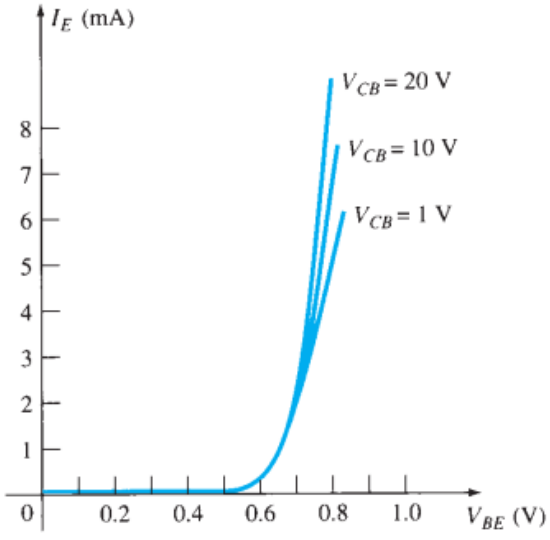
When free electrons from emitter move to the base the free electrons and the free holes combine with each other but since the base is very thin only some free electrons gets combined with the holes and the majority of the electrons are attracted towards the collector because of the positive terminal voltage connected to the collector. Thus the current flow through the output terminal.

Thus the emitter current is the sum of the base current and the collector current.

$I_E = I_B + I_C$

In common base configuration, the input impedance is low and the output impedance is high and the overall power gain is low when compared with other configuration

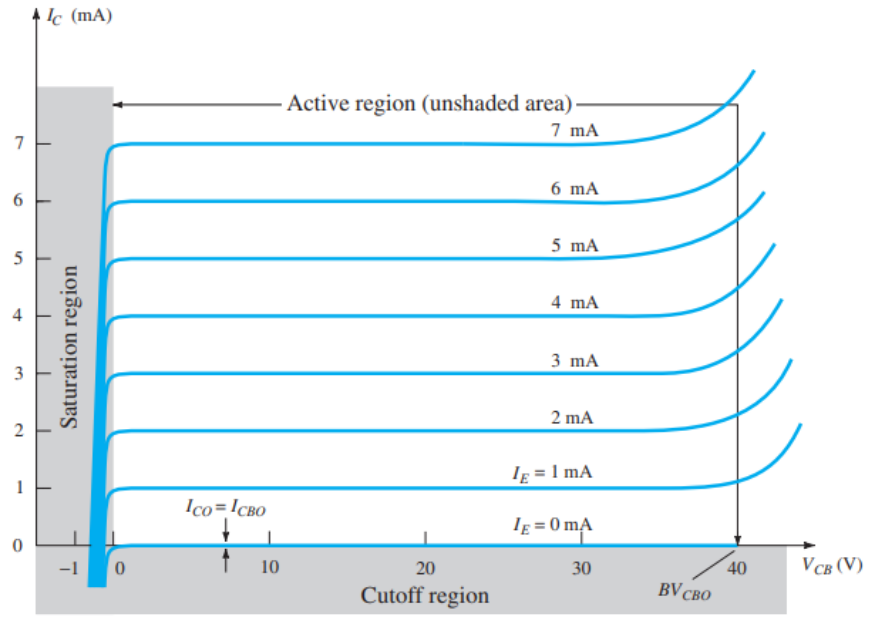
Input or driving point characteristics:



Input characteristics are the relationship between the input current and input voltage with constant output voltage. In common base configuration input current is emitter current I_E and the input voltage is base emitter voltage V_{BE}. The curve is plotted between I_E and V_{BE} keeping V_{CB} as constant. The V_{BE} is increased keeping V_{CB} constant, initially at zero and the input current I_E is noted, similarly the V_{CB} value is increased and kept constant and V_{BE} is increased and the input current I_E is noted.

Input side is forward biased so the input resistance is small so for a small increase in V_{BE} there is rapid increase in the emitter current I_E. As the output voltage V_{CB} is increased the width of the depletion layer between emitter base decreases and the cut in voltage is reduced so the curve drifts to the left side.

Output or collector characteristics:



Output characteristics are the relationship between output current I_C and output voltage V_{CB} keeping input current I_E constant. When the input current I_E is zero it is in cut off region. In saturation region both emitter base junction and collector base junction are forward biased.

In active region I_E is gradually increased and kept constant and output voltage V_{CB} is increased further and the output current I_C almost remains constant. So in active region curve is almost flat. Output voltage causes only a very little change in output current.

In the active region the base–emitter junction is forward-biased, whereas the collector–base junction is reverse-biased.

$$I_C \approx I_E$$

In the cutoff region the base–emitter and collector–base junctions of a transistor are both reverse-biased.

In the saturation region the base–emitter and collector–base junctions are forward-biased.

Alpha (α):

DC Mode In the dc mode the levels of I_C and I_E due to the majority carriers are related by a quantity called alpha and defined by the following equation:

$$\alpha_{dc} = \frac{I_C}{I_E}$$

Since alpha is defined solely for the majority carriers, above equation can be written as

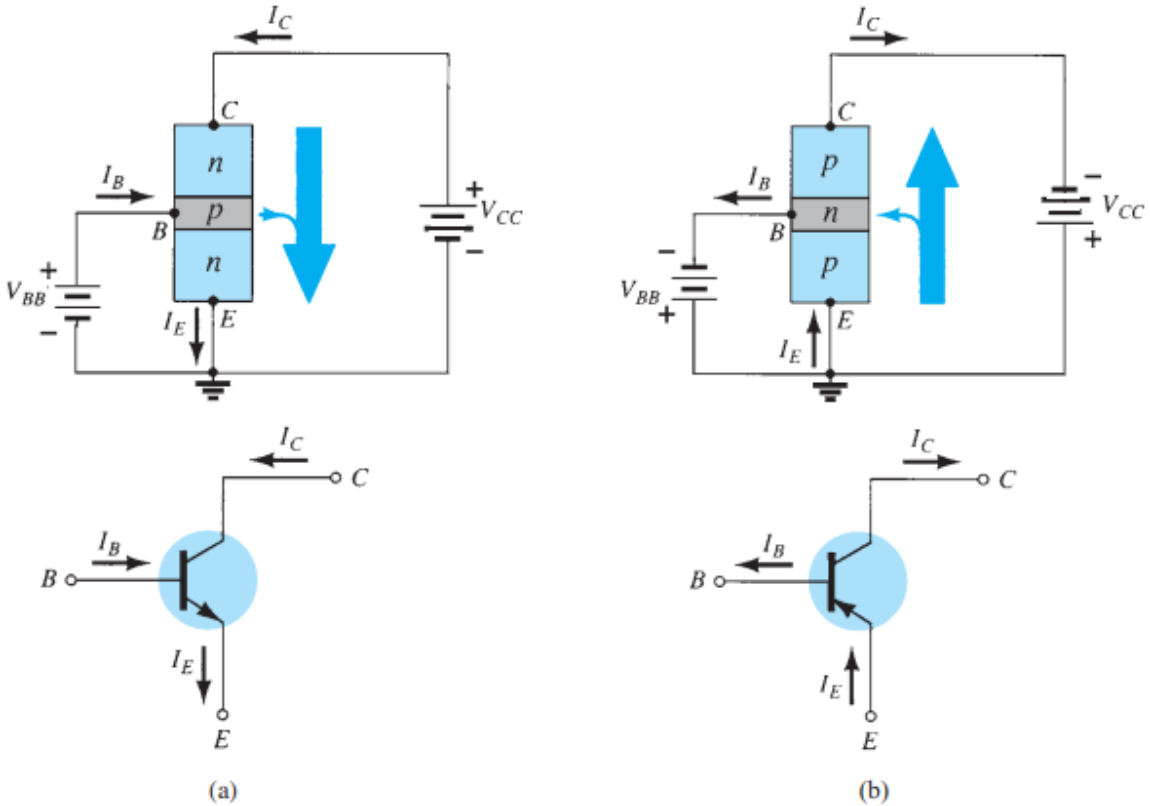
$$I_C = \alpha I_E + I_{CBO}$$

AC Mode For ac situations where the point of operation moves on the characteristic curve, an ac alpha is defined by

$$\alpha_{ac} = \left.\frac{\Delta I_C}{\Delta I_E}\right|_{V_{CB}=\text{constant}}$$

Common Emitter Configuration:

In common emitter configuration, base is the input terminal, collector is the output terminal and emitter is the common terminal for both input and output. That means the base terminal and common emitter terminal are known as input terminals whereas collector terminal and common emitter terminal are known as output terminals.



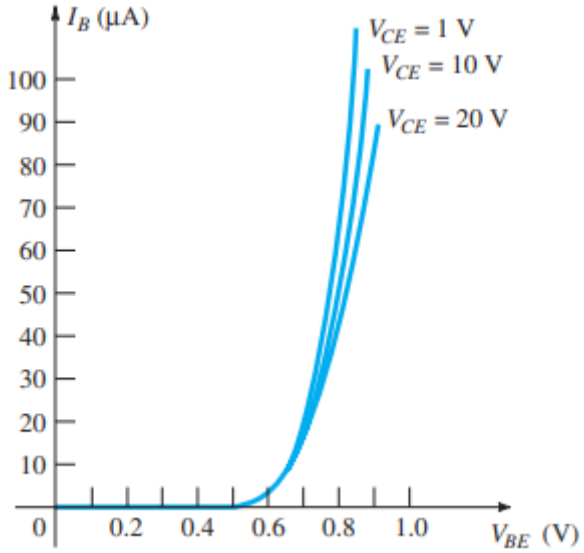
In common emitter configuration, the emitter terminal is grounded so the common emitter configuration is also known as grounded emitter configuration. Sometimes common emitter configuration is also referred to as CE configuration, common emitter amplifier, or CE amplifier. The common emitter (CE) configuration is the most widely used transistor configuration.

The common emitter (CE) amplifiers are used when large current gain is needed.
The input signal is applied between the base and emitter terminals while the output signal is taken between the collector and emitter terminals. Thus, the emitter terminal of a transistor is common for both input and output and hence it is named as common emitter configuration.
The supply voltage between base and emitter is denoted by V_{BE} while the supply voltage between collector and emitter is denoted by V_{CE}.

In common emitter (CE) configuration, input current or base current is denoted by I_B and output current or collector current is denoted by I_C.

The common emitter amplifier has medium input and output impedance levels. So the current gain and voltage gain of the common emitter amplifier is medium. However, the power gain is high.

Input characteristics:



The input characteristics are a plot of the input current (I_B) versus the input voltage (V_{BE}) for a range of values of output voltage (V_{CE}). To determine the input characteristics, the output voltage V_{CE} is kept constant at zero volts and the input voltage V_{BE} is increased from zero volts to different voltage levels. For each voltage level of input voltage (V_{BE}), the corresponding input current (I_B) is recorded.

A curve is then drawn between input current I_B and input voltage V_{BE} at constant output voltage V_{CE} (0 volts).

Next, the output voltage (V_{CE}) is increased from zero volts to certain voltage level (10 volts) and the output voltage (V_{CE}) is kept constant at 10 volts. While increasing the output voltage (V_{CE}), the input voltage (V_{BE}) is kept constant at zero volts. After we kept the output voltage (V_{CE}) constant at 10 volts, the input voltage V_{BE} is increased from zero volts to different voltage levels. For each voltage level of input voltage (V_{BE}), the corresponding input current (I_B) is recorded.

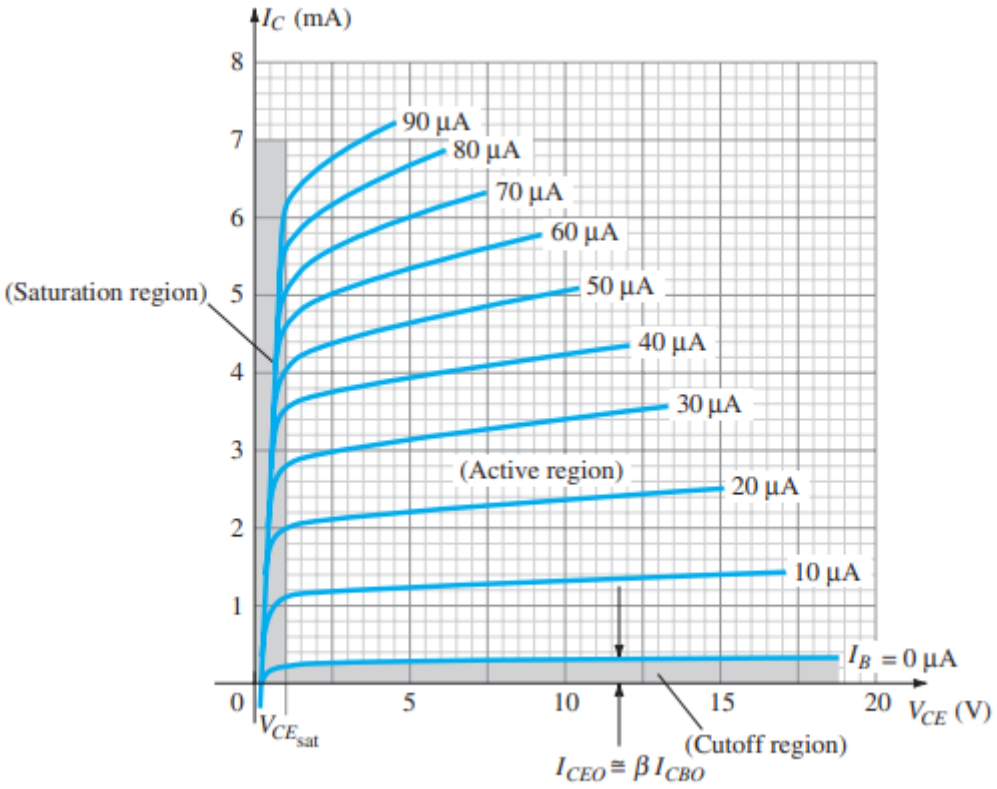
A curve is then drawn between input current I_B and input voltage V_{BE} at constant output voltage V_{CE} (10 volts). This process is repeated for higher fixed values of output voltage (V_{CE}).

When output voltage (V_{CE}) is at zero volts and emitter-base junction is forward biased by input voltage (V_{BE}), the emitter-base junction acts like a normal p-n junction diode. So the input characteristics of the CE configuration is same as the characteristics of a normal pn junction diode.

The cut in voltage of a silicon transistor is 0.7 volts and germanium transistor is 0.3 volts. In our case, it is a silicon transistor. So from the above graph, we can see that after 0.7 volts, a small increase in input voltage (V_{BE}) will rapidly increases the input current (I_B).

In common emitter (CE) configuration, the input current (I_B) is very small as compared to the input current (I_E) in common base (CB) configuration. The input current in CE configuration is measured in microamperes (μA) whereas the input current in CB configuration is measured in milliamperes (mA). From the above characteristics, we can see that for higher fixed values of output voltage V_{CE}, the curve shifts to the right side. This is because for higher fixed values of output voltage, the cut in voltage is increased above 0.7 volts. Therefore, to overcome this cut in voltage, more input voltage V_{BE} is needed than previous case.

Output characteristics:



For the common-emitter configuration the output characteristics are a plot of the output

current (I_C) versus output voltage (V_{CE}) for a range of values of input current (I_B). To determine the output characteristics, the input current or base current I_B is kept constant at 0 μA and the output voltage V_{CE} is increased from zero volts to different voltage levels. For each level of output voltage, the corresponding output current (I_C) is recorded.

A curve is then drawn between output current I_C and output voltage V_{CE} at constant input current I_B (0 μA).

When the base current or input current I_B = 0 μA, the transistor operates in the cut-off region. In this region, both junctions are reverse biased.

Next, the input current (I_B) is increased from 0 μA to 20 μA by adjusting the input voltage (V_{BE}). The input current (I_B) is kept constant at 20 μA.

While increasing the input current (I_B), the output voltage (V_{CE}) is kept constant at 0 volts.

After we kept the input current (I_B) constant at 20 μA, the output voltage (V_{CE}) is increased from zero volts to different voltage levels. For each voltage level of output voltage (V_{CE}), the corresponding output current (I_C) is recorded.

A curve is then drawn between output current I_C and output voltage V_{CE} at constant input current I_B (20 μA). This region is known as the active region of a transistor. In this region, emitter-base junction is forward biased and the collector-base junction is reverse biased.

This steps are repeated for higher fixed values of input current I_B (I.e. 40 μA, 60 μA, 80 μA and so on).

When output voltage V_{CE} is reduced to a small value (0.2 V), the collector-base junction becomes forward biased. This is because the output voltage V_{CE} has less effect on collector-base junction than input voltage V_{BE}.

As we know that the emitter-base junction is already forward biased. Therefore, when both the junctions are forward biased, the transistor operates in the saturation region. In this region, a small increase in output voltage V_{CE} will rapidly increases the output current I_C.

Beta (β)

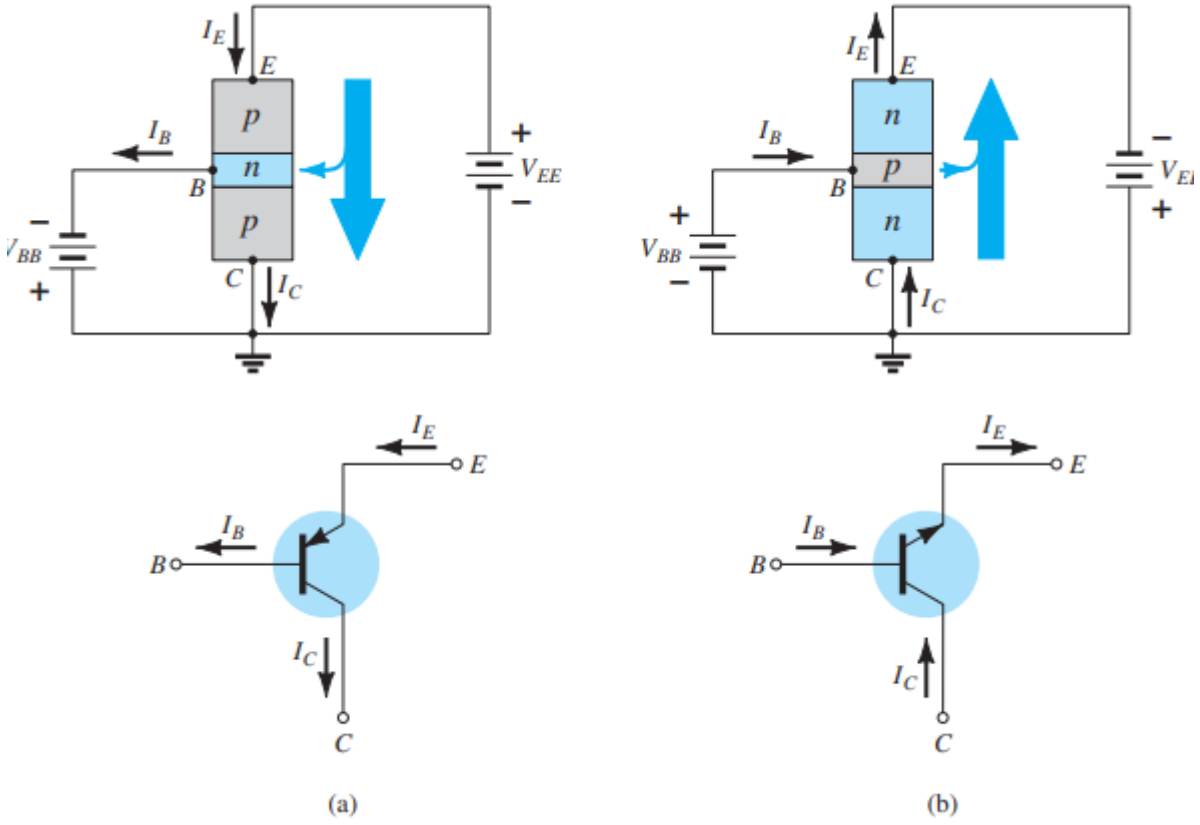
DC Mode In the dc mode the levels of I_C and I_B are related by a quantity called beta and defined by the following equation:

$$\beta_{dc} = \frac{I_C}{I_B}$$

AC Mode For ac situations an ac beta is defined as follows:

$$\beta_{ac} = \left.\frac{\Delta I_C}{\Delta I_B}\right|_{V_{CE}=\text{constant}}$$

Common Collector Configuration:



The common-collector configuration is used primarily for impedance-matching purposes since it has a high input impedance and low output impedance, opposite to that of the common-base and common emitter configurations.

In this configuration, the base terminal of the transistor serves as the input, the emitter terminal is the output and the collector terminal is common for both input and output. Hence, it is named as common collector configuration. The input is applied between the base and collector while the output is taken from the emitter and collector.

Sometimes common collector configuration is also referred to as emitter follower, voltage follower, common collector amplifier, CC amplifier, or CC configuration. This configuration is mostly used as a voltage buffer.

The input supply voltage between base and collector is denoted by V_{BC} while the output voltage between emitter and collector is denoted by V_{EC}.

In this configuration, input current or base current is denoted by I_B and output current or emitter current is denoted by I_E. The common collector amplifier has high input impedance and low output impedance. It has low voltage gain and high current gain.

The power gain of the common collector amplifier is medium.

The output characteristics of the common-collector configuration are the same as for the common-emitter configuration. For the common-collector configuration the output characteristics are a plot of I_E versus V_{CE} for a range of values of I_B . The input current,

therefore, is the same for both the common emitter and common-collector characteristics.

DC Biasing-

Basic relationships for a transistor:

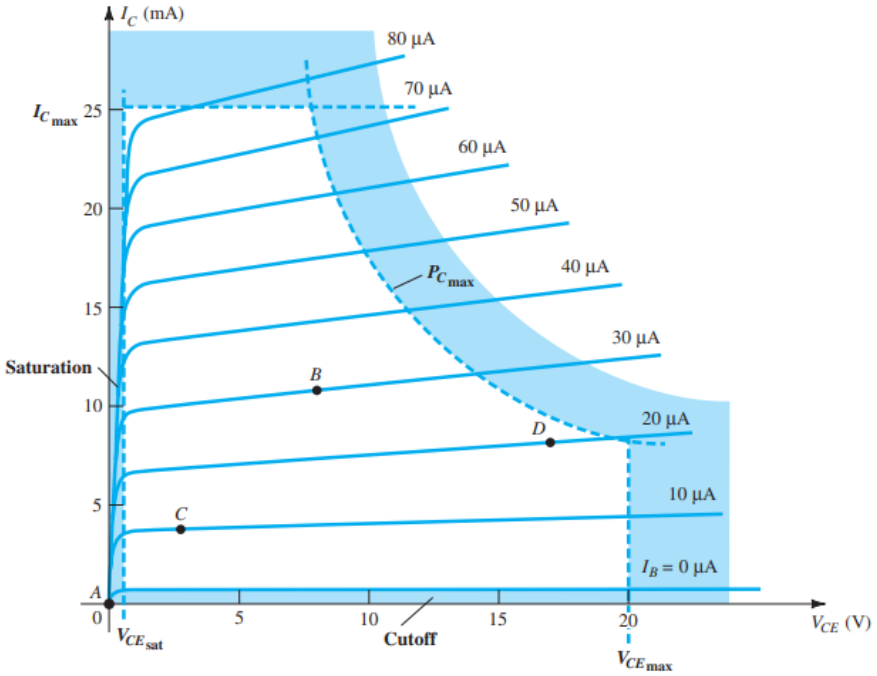
$$V_{BE} \cong 0.7\ \text{V}$$

$$I_E = (\beta + 1)I_B \cong I_C$$

$$I_C = \beta I_B$$

Operating point

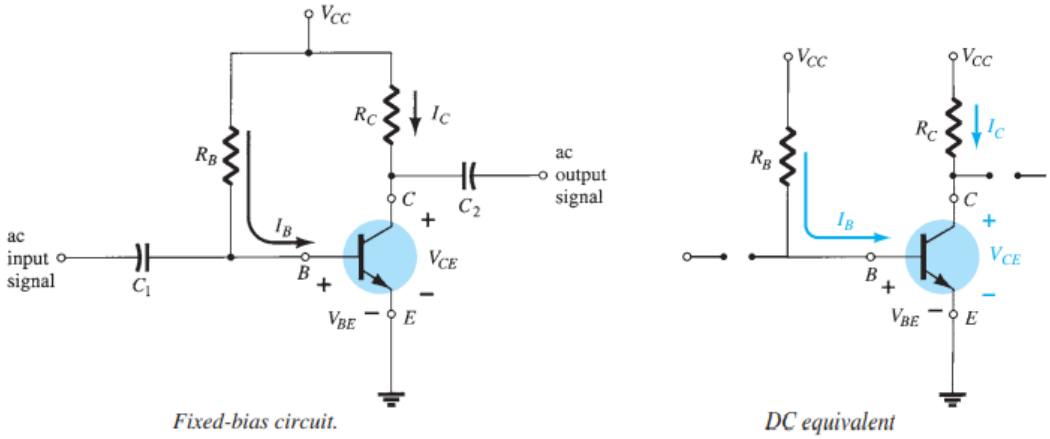
For transistor amplifiers the resulting dc current and voltage establish an operating point on the characteristics that define the region that will be employed for amplification of the applied signal. Because the operating point is a fixed point on the characteristics, it is also called the quiescent point (abbreviated Q -point). By definition, quiescent means quiet, still, inactive.



For the BJT to be biased in its linear or active operating region the following must be true:
1. The base–emitter junction must be forward-biased (p-region voltage more positive), with a resulting forward-bias voltage of about 0.6 V to 0.7 V.
2. The base–collector junction must be reverse-biased (n-region more positive), with the reverse-bias voltage being any value within the maximum limits of the device.
Operation in the cutoff, saturation, and linear regions of the BJT characteristic are provided as follows:

1. Linear-region operation:
Base–emitter junction forward-biased
Base–collector junction reverse-biased
2. Cutoff-region operation:
Base–emitter junction reverse-biased
Base–collector junction reverse-biased
3. Saturation-region operation:
Base–emitter junction forward-biased
Base–collector junction forward-biased

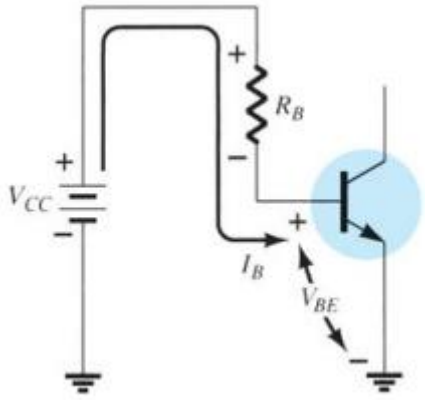
Fixed Bias configuration



Fixed-bias circuit. *DC equivalent*

The circuit shown is called as a "fixed base bias circuit", because the transistors base current, I_B remains constant for given values of Vcc, and therefore the transistors operating point must also remain fixed. This two resistor biasing network is used to establish the initial operating region of the transistor using a fixed current bias.

Forward Bias of Base–Emitter



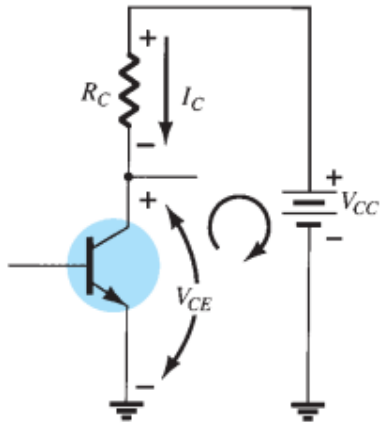
Writing Kirchhoff's voltage equation in the clockwise direction for the loop, we obtain

$+V_{CC} - I_B R_B - V_{BE} = 0$ ----------1

Solving the equation for the current I B results in the following:

$$I_B = \frac{V_{CC} - V_{BE}}{R_B}$$

Collector–Emitter Loop



The magnitude of the collector current is related directly to I_B through

$$I_C = \beta I_B$$

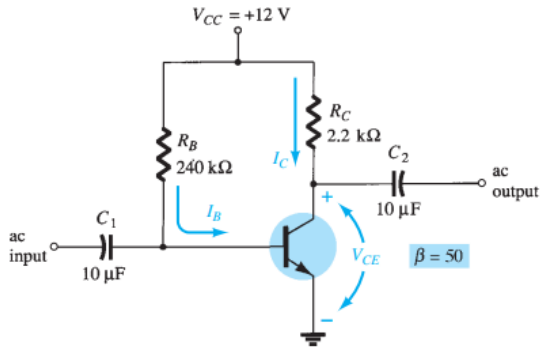
Applying Kirchhoff's voltage law in the clockwise direction around the indicated closed Loop

$V_{CE} + I_C R_C - V_{CC} = 0$

$$V_{CE} = V_{CC} - I_C R_C$$

Problem:

1. Determine the following for the fixed-bias configuration of following fig.
   a. I_{BQ} and I_{CQ}.
   b. V_{CEQ}.
   c. V_B and V_C
   d. V_{BC}



Solution:

a.

$$I_{B_Q} = \frac{V_{CC} - V_{BE}}{R_B} = \frac{12\text{ V} - 0.7\text{ V}}{240\text{ k}\Omega} = \mathbf{47.08\ \mu A}$$

$$I_{C_Q} = \beta I_{BQ} = (50)(47.08\ \mu\text{A}) = \mathbf{2.35\ mA}$$

b.

$$V_{CE_Q} = V_{CC} - I_C R_C$$
$$= 12\text{ V} - (2.35\text{ mA})(2.2\text{ k}\Omega)$$
$$= \mathbf{6.83\ V}$$

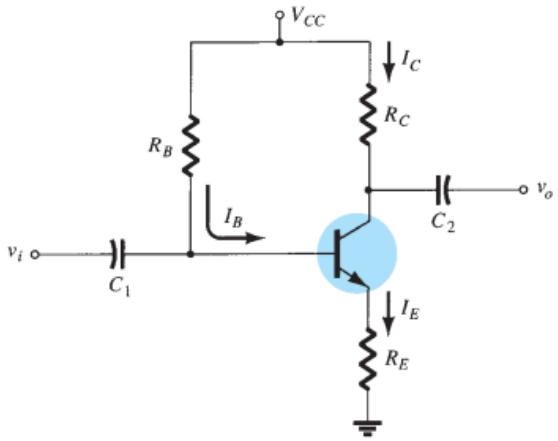
c.

$$V_B = V_{BE} = \mathbf{0.7\ V}$$
$$V_C = V_{CE} = \mathbf{6.83\ V}$$

d.

$$V_{BC} = V_B - V_C = 0.7\text{ V} - 6.83\text{ V}$$
$$= \mathbf{-6.13\ V}$$

Emitter Bias configuration



- The circuit contains an emitter resistor to improve the stability level over that of the fixed-bias configuration. The more stable a configuration, the less its response will change due to undesirable changes in temperature and parameter variations.

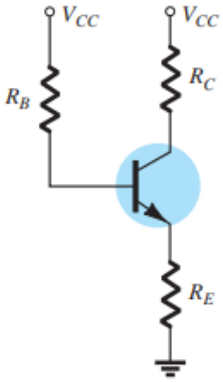


Fig. DC equivalent

Base–Emitter Loop

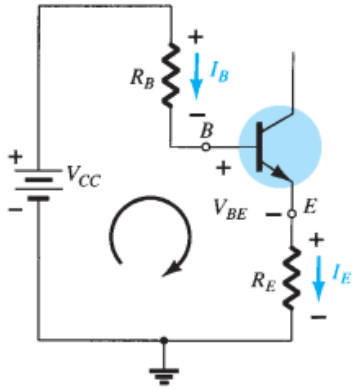


Fig. Base–emitter loop.

Writing Kirchhoff's voltage law around the indicated loop in the clockwise direction results in the following equation:

$$+V_{CC} - I_B R_B - V_{BE} - I_E R_E = 0$$

$$I_E = (\beta + 1) I_B$$

Substituting for I_E

$$V_{CC} - I_B R_B - V_{BE} - (\beta + I) I_B R_E = 0$$

Grouping terms then provides the following:

$$-I_B(R_B + (\beta + 1)R_E) + V_{CC} - V_{BE} = 0$$

Multiplying through by (-1), we have

$$I_B(R_B + (\beta + 1)R_E) - V_{CC} + V_{BE} = 0$$

with

$$I_B(R_B + (\beta + 1)R_E) = V_{CC} - V_{BE}$$

and solving for I_B gives

$$\boxed{I_B = \frac{V_{CC} - V_{BE}}{R_B + (\beta + 1)R_E}}$$

Collector–Emitter Loop

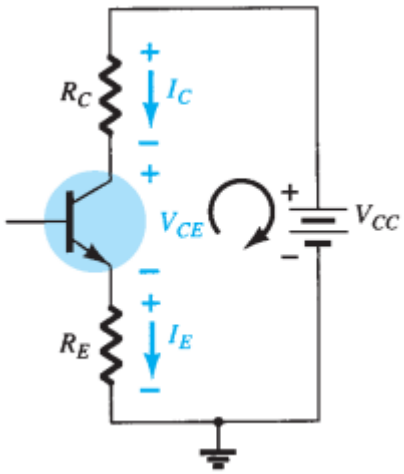


Fig. Collector–emitter loop.

Writing Kirchhoff's voltage law for the indicated loop in the clockwise direction results in

$$+I_E R_E + V_{CE} + I_C R_C - V_{CC} = 0$$

Substituting $I_E \cong I_C$ and grouping terms gives

$$V_{CE} - V_{CC} + I_C(R_C + R_E) = 0$$

and

$$\boxed{V_{CE} = V_{CC} - I_C(R_C + R_E)}$$

The single-subscript voltage V_E is the voltage from emitter to ground and is determined by

$$\boxed{V_E = I_E R_E}$$

whereas the voltage from collector to ground can be determined from

$$V_{CE} = V_C - V_E$$

and

$$\boxed{V_C = V_{CE} + V_E}$$

or

$$\boxed{V_C = V_{CC} - I_C R_C}$$

The voltage at the base with respect to ground

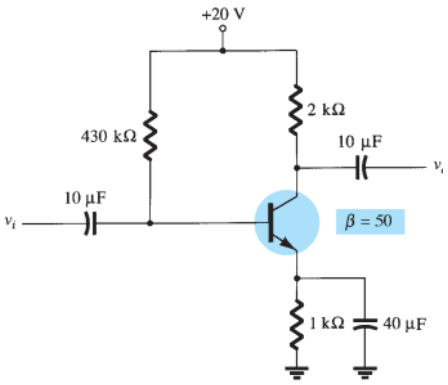
$$\boxed{V_B = V_{CC} - I_B R_B}$$

or

$$\boxed{V_B = V_{BE} + V_E}$$

Problem:

1. For the emitter -bias configuration of following fig. determine:
   a. I_B
   b. I_C
   c. V_{CE}
   d. V_C
   e. V_E
   f. V_B
   g. V_{BC}



Solution:

a.

$$I_B = \frac{V_{CC} - V_{BE}}{R_B + (\beta + 1)R_E} = \frac{20\text{ V} - 0.7\text{ V}}{430\text{ k}\Omega + (51)(1\text{ k}\Omega)}$$

$$= \frac{19.3\text{ V}}{481\text{ k}\Omega} = \mathbf{40.1\ \mu A}$$

b.

$$I_C = \beta I_B$$

$$= (50)(40.1\ \mu\text{A})$$

$$\cong \mathbf{2.01\ mA}$$

c.

$$V_{CE} = V_{CC} - I_C(R_C + R_E)$$

$$= 20\text{ V} - (2.01\text{ mA})(2\text{ k}\Omega + 1\text{ k}\Omega) = 20\text{ V} - 6.03\text{ V}$$

$$= \mathbf{13.97\ V}$$

d. $V_C = V_{CC} - I_C R_C$

$$= 20\text{ V} - (2.01\text{ mA})(2\text{ k}\Omega) = 20\text{ V} - 4.02\text{ V}$$

$$= \mathbf{15.98\ V}$$

e. $V_E = V_C - V_{CE}$

$$= 15.98\text{ V} - 13.97\text{ V}$$

$$= \mathbf{2.01\ V}$$

f. $V_B = V_{BE} + V_E$

$$= 0.7\text{ V} + 2.01\text{ V}$$

$$= \mathbf{2.71\ V}$$

g. $V_{BC} = V_B - V_C$
$= 2.71\text{ V} - 15.98\text{ V}$
$= \mathbf{-13.27\text{ V}}$ (reverse-biased as required)

Voltage-Divider Bias Configuration

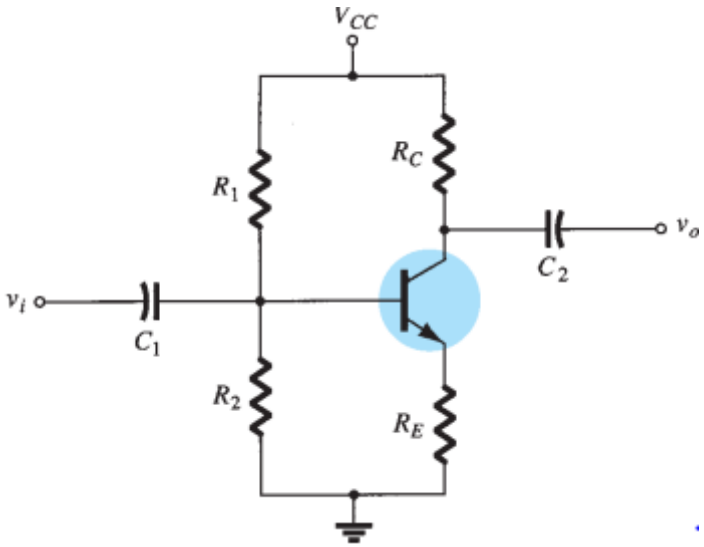


Fig. Voltage-divider bias configuration

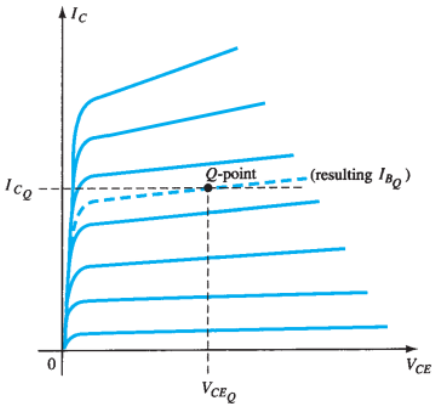


Fig. Defining the Q-point for the voltage-divider bias configuration

Exact Analysis:

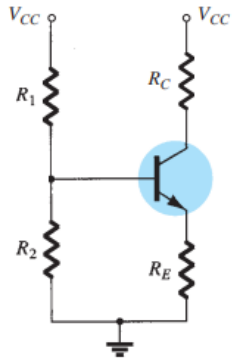


Fig. DC components of the voltage divider configuration

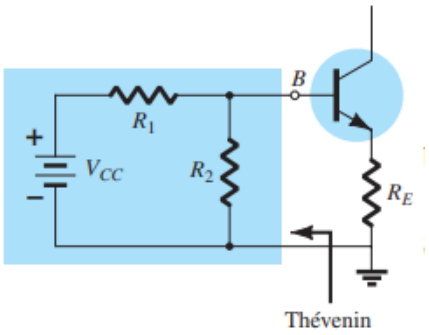


Fig. Redrawing the input side of the network

The Thevenin equivalent network for the network to the left of the base terminal can be found as follows:

R_{Th} The voltage source is replaced by a short-circuit equivalent

$$R_{Th} = R_1 \| R_2$$

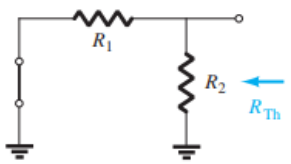


Fig. Determining R_{Th}

E_{Th} The voltage source V_{CC} is returned to the network and the open-circuit Thevenin voltage of following Fig. determined as follows:

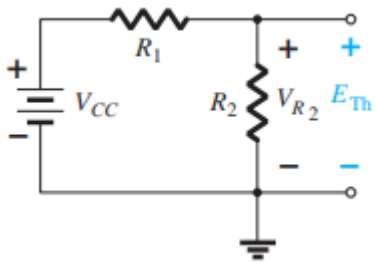


Fig. Determining E_{Th}

Applying the voltage-divider rule gives

$$E_{Th} = V_{R_2} = \frac{R_2 V_{CC}}{R_1 + R_2}$$

The Thevenin network is then redrawn as shown in Fig., and I_{BQ} can be determined by first applying Kirchhoff's voltage law in the clockwise direction for the loop indicated:

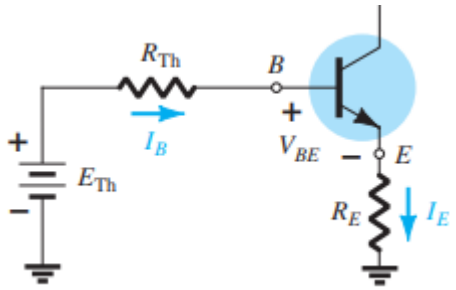


Fig. Inserting the Thevenin equivalent circuit

$$E_{Th} - I_B R_{Th} - V_{BE} - I_E R_E = 0$$

Substituting $I_E = (\beta + 1)I_B$ and solving for I_B yields

$$I_B = \frac{E_{Th} - V_{BE}}{R_{Th} + (\beta + 1)R_E}$$

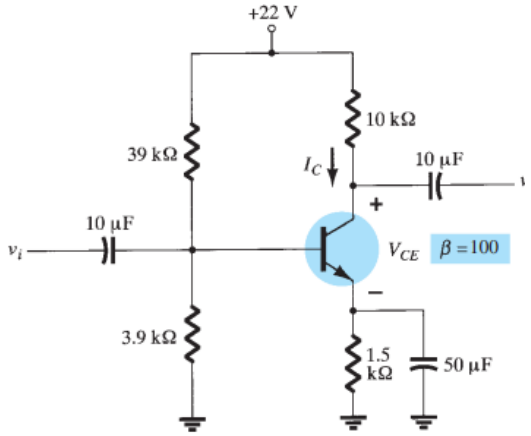
Once I_B is known, the remaining quantities of the network can be found in the same manner as developed for the emitter-bias configuration. That is,

$$V_{CE} = V_{CC} - I_C(R_C + R_E)$$

Note: The remaining equations for V_E, V_C, and V_B are also the same as obtained for the emitter-bias configuration.

Problem:

1. Determine the dc bias voltage V_{CE} and the current I_C for the voltagedivider configuration of the following Fig.



Solution:

$$R_{Th} = R_1 \| R_2$$
$$= \frac{(39\ \text{k}\Omega)(3.9\ \text{k}\Omega)}{39\ \text{k}\Omega + 3.9\ \text{k}\Omega} = 3.55\ \text{k}\Omega$$
$$E_{Th} = \frac{R_2 V_{CC}}{R_1 + R_2}$$
$$= \frac{(3.9\ \text{k}\Omega)(22\ \text{V})}{39\ \text{k}\Omega + 3.9\ \text{k}\Omega} = 2\ \text{V}$$
$$I_B = \frac{E_{Th} - V_{BE}}{R_{Th} + (\beta + 1)R_E}$$
$$= \frac{2\ \text{V} - 0.7\ \text{V}}{3.55\ \text{k}\Omega + (101)(1.5\ \text{k}\Omega)} = \frac{1.3\ \text{V}}{3.55\ \text{k}\Omega + 151.5\ \text{k}\Omega}$$
$$= 8.38\ \mu\text{A}$$
$$I_C = \beta I_B$$
$$= (100)(8.38\ \mu\text{A})$$
$$= \mathbf{0.84\ mA}$$

$$V_{CE} = V_{CC} - I_C(R_C + R_E)$$
$$= 22\ \text{V} - (0.84\ \text{mA})(10\ \text{k}\Omega + 1.5\ \text{k}\Omega)$$
$$= 22\ \text{V} - 9.66\ \text{V}$$
$$= \mathbf{12.34\ V}$$

Approximate Analysis:

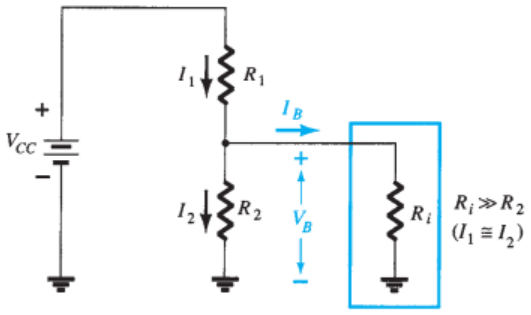


Fig. Partial-bias circuit for calculating the approximate base voltage V_B

determined using the voltage-divider rule (hence the name for the configuration). That is,

$$\boxed{V_B = \frac{R_2 V_{CC}}{R_1 + R_2}}$$

Because $R_i = (\beta + 1)R_E \cong \beta R_E$ the condition that will define whether the approximate approach can be applied is

$$\boxed{\beta R_E \geq 10R_2}$$

In other words, if β times the value of R_E is at least 10 times the value of R_2, the approximate approach can be applied with a high degree of accuracy.

Once V_B is determined, the level of V_E can be calculated from

$$\boxed{V_E = V_B - V_{BE}}$$

and the emitter current can be determined from

$$\boxed{I_E = \frac{V_E}{R_E}}$$

and

$$\boxed{I_{C_Q} \cong I_E}$$

The collector-to-emitter voltage is determined by

$$V_{CE} = V_{CC} - I_C R_C - I_E R_E$$

but because $I_E \cong I_C$,

$$\boxed{V_{CE_Q} = V_{CC} - I_C(R_C + R_E)}$$